AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM340
Natural Language Processing

Midterm Examination (MTA) - Makeup
Spring Semester 2024/2025

Date: 24 /Apr/2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1

Multiple Choice Questions **(10 marks)**

Question 1, (10 marks, 1 mark for each question)
Answer the following MCQ's by selecting **the best** answer for each question. Write your **answer letter** in the answer sheet.

1. **Which component of linguistic levels focuses on the structural relationships between words?**

   A) Phonetics
   B) Morphology
   C) Syntax
   D) Discourse

2. **Which task is NOT part of the core tasks of Natural Language Processing?**

   A) Image Recognition
   B) Text Classification
   C) Machine Translation
   D) Summarization

3. **Which paradigm is commonly associated with the earliest NLP systems?**

   A) Deep Learning NLP
   B) Rule-based NLP
   C) Statistical NLP
   D) Neural NLP

4. **In word tokenization, what does a 'lemma' represent?**

   A) The exact form of a word
   B) The base form or root of a word
   C) A punctuation mark
   D) A number in the text

5. **In regular expressions, what does the pattern \S match?**

   A) A digit
   B) A non-digit character
   C) A whitespace character
   D) A non-whitespace character

6. **Why is log space often used when computing N-gram probabilities?**

   A) To handle underflow when multiplying small probabilities
   B) To increase the training speed
   C) To generate more coherent sentences
   D) To avoid zero probabilities

7. **How does interpolation improve N-gram models?**

   A) It simplifies the model by removing rare N-grams
   B) It combines probabilities from different N-gram models
   C) It ignores stop words during training
   D) It increases the vocabulary size

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8. **What is the main limitation of using sigmoid function in logistic regression for multi-class classification?**

   A) It requires negative weights
   B) It leads to overfitting
   C) It cannot handle more than two classes
   D) It is computationally expensive

9. **Which of the following is a key assumption in Naive Bayes?**

   A) All features are independent given the class
   B) Words occur in order
   C) The training data is unlabeled
   D) It cannot handle categorical data

10. **What issue does multiplying many small probabilities lead to in Naive Bayes?**

    A) Overfitting
    B) Floating-point underflow
    C) Increased accuracy
    D) Infinite probability values

PART 2

Short Essay Questions **(10 marks)**

Question 2, (10 marks)
Answer the following questions:

a) List the 4 steps of the NLP pipeline and briefly describe each step?
b) Describe how the Markov assumption simplifies language modeling?
c) What is Laplace smoothing, and why is it used in Naive Bayes?

PART 3

Short problems/questions **(40 marks)**

Question 3, (10 marks)
Find the minimum number of operations (insertions, deletions, and substitutions) required to transform word **book** into **back**. Using the *Levenshtein Minimum Edit Distance algorithm* (insertions and deletions cost 1, while substitutions cost 2):

a) Complete the distance matrix for the given words.

4	k					
3	o					
2	o					
1	b					
0	#					
		#	b	a	c	k
		0	1	2	3	4

b) Using the distance matrix, what is the minimum edit distance value.

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Question 4, (10 marks)
Suppose you have a deterministic language L = {cat, dog, mouse, bird} where each word in a sentence is chosen from L (branching factor = 4). The training data provides the following probabilities:

p(cat) = 0.6 **p(dog) = 0.2** **p(mouse) = 0.1** **p(bird) = 0.1**

For the test set S = "**cat dog cat mouse**":

a) Compute the probability of **S** for this model.
b) Calculate the perplexity of the sentence **S** for this model.

Question 5, (10 marks)
Suppose that you have a text classifier and test it on a dataset containing 150 examples distributed across three classes:

- Science: 70 instances
- News: 50 instances
- Sports: 30 instances

And the confusion matrix was:

		True Class		
		Science	News	Sports
Predicted Class	Science	55	5	3
	News	10	40	5
	Sports	5	5	22

a) According to the confusion matrix, compute the **Recall** for each class?

b) According to the confusion matrix, Compute the **Microaveraging Recall**?

Question 6, (10 marks)
Assume the following likelihoods for each word being part of a ***Positive vs. Negative*** Restaurant Review:

Word	Positive	Negative
food	0.25	0.05
tasty	0.20	0.03
slow	0.05	0.15
service	0.10	0.10
staff	0.08	0.12

Suppose that the **prior probabilities** are **0.35** for ***Positive*** class and **0.65** for ***Negative*** class.

And given the document d = "***food tasty slow service***", Consider the ***naive Bayes classifier***, **what class** will Naive Bayes assign to the document d? **Show** the **steps** and **calculations** required to decide the correct class.

End of questions

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